

C1: Measure Identification

H&K Stages S0-S3 — final codebook (v0.7.0) and development history

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C1 implements R&R Step 2 (Measure Identification): scanning a document chunk and deciding whether any portion of it substantively discusses a specific fiscal measure. The current deliverable is **v0.7.0**, accepted for deployment at S3 iteration 33. This notebook has two parts: the full final-iteration battery (mimicking the Halterman & Keith presentables) and the development history that produced it. The cross-codebook rollup lives in `iteration_summary.qmd`.

Part 1: The Final Codebook (v0.7.0)

S0: Codebook and Prompt

C1 is a binary classification task with two classes:

- **FISCAL_MEASURE**: any portion of the chunk substantively discusses a specific enacted, proposed, or under-consideration fiscal action.
- **NOT_FISCAL_MEASURE**: everything else (general economic commentary, aggregates, administrative procedure, bare name-drops).

Under v0.7.0 the positive class additionally returns a structured `measures[]` array (one entry per distinct act, with a per-measure country tag and three relevance flags), which is what downstream codebooks consume.

Table 1: C1 codebook structure (live `c1_codebook`).

Field	Value
Name	C1: Fiscal Measure Identification
Version	0.7.0
Classes	2
Clarifications per class	6, 4
Neg. clarifications per class	5, 2

Evaluation negatives

The zero-shot evaluation draws stratified hard negatives from `us_body`, designed to test the most common confusion patterns.

Table 2: Stratified hard-negative pool for evaluation.

Type	Count	%
economic_trends	90	30.0
extensions	60	20.0
proposals	120	40.0
withholding	30	10.0

Table 3: Evaluation data sources.

Source	Count
Positive (aligned_data passages)	368
Negative (stratified hard negatives)	300
Total	668

The full prompt

This is the assembled system prompt the model saw for the US evaluation run, with the `{country}` / `{country_iso}` tokens resolved (here, the United States). At deployment the same template is re-rendered per country.

i Prompt excerpt from C1

You are an expert in fiscal policy analysis classifying chunks from official government economic and budget documents. Your task is to scan the entire chunk and determine whether **any** portion of it substantively discusses a specific fiscal measure – enacted, proposed, or under consideration – that changes or would change fiscal liabilities or obligations. Chunks from government documents typically contain extended economic analysis, forecasts, and policy discussion – a chunk should be classified as FISCAL_MEASURE if it contains even one substantive discussion of a specific fiscal measure, regardless of what the rest of the chunk discusses. Read the class definitions carefully and classify the chunk using only the provided labels.

The chunk you are analysing comes from a corpus published by the government of **United States** (ISO code: `US`). Some chunks discuss fiscal measures enacted by other countries (foreign comparators, supranational bodies); these still qualify as FISCAL_MEASURE because they describe fiscal policy, but their country-of-enactment is `OTHER`, not `US`. The country-of-enactment tag is applied per measure, since a single chunk may discuss measures from multiple countries.

Class Definitions

FISCAL_MEASURE

****Definition:**** A chunk that contains, in any portion of its text, a substantive discussion of a specific legislative or executive action – whether enacted, proposed, or under consideration – that changes or would change fiscal liabilities or obligations.

****Inclusion criteria:****

- The passage describes a change in fiscal liabilities from taxes of any type – income taxes, payroll and social insurance taxes, consumption and excise taxes, investment incentives, and so on
- The action may be enacted into law, implemented by executive or administrative action (such as executive orders, ministerial decrees, regulatory changes, or official policy directives), proposed in legislation, recommended by officials or advisory bodies, or under consideration; C1 captures all

substantive discussion of specific fiscal measures regardless of legislative status

- Retrospective references that³ name a specific act and provide substantive detail about its provisions – such as rate changes, affected populations, or revenue effects – should be classified as FISCAL_MEASURE, even if the passage was written years after enactment

S1: Behavioral Tests (H&K Figure 3)

S1 checks that the model emits legal labels, can recover the class definitions from memory, and is insensitive to class ordering. Test III (in-context example recovery) is **skipped** for C1: v0.7.0 carries no in-codebook examples.

Figure 1: S1 behavioral tests for C1 v0.7.0. Tests I/II shown as percent correct; Test IV shown as Fleiss' kappa with Landis (1977) reference bands.

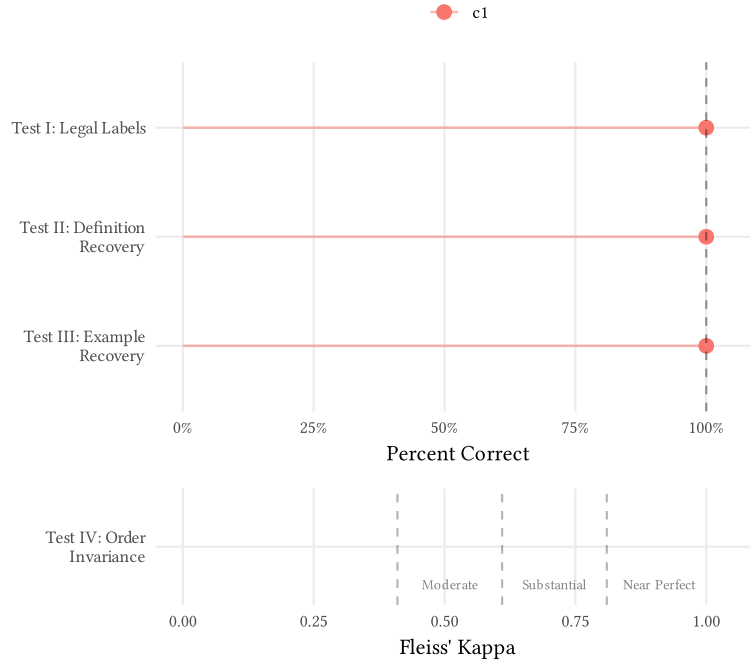


Table 4: S1 results for the latest C1 iteration.

Test	Value	Threshold	Status
I_legal_outputs	100%	$\geq 100\%$	Pass
II_definition_recovery	100%	$\geq 100\%$	Pass
IV_order_invariance	0.0% change	$< 5\%$	Pass

S2: Zero-Shot Evaluation

Tier 1 chunks carry verbatim R&R passages (high-confidence labels); Tier 2 chunks are name-matched via the IKA heuristic (noisier). Tier 1 recall is the metric we trust most.

Table 5: S2 zero-shot metrics with 95% bootstrap CIs (latest C1 iteration).

Metric	Estimate	Target	Status
combined_recall	0.800 [0.771, 0.829]	≥ 0.90	Fail
tier1_recall	0.882 [0.803, 0.955]	≥ 0.95	Fail
precision	0.947 [0.929, 0.964]	≥ 0.70	Pass
tier2_recall	0.792 [0.762, 0.822]	—	—
f1	0.867 [0.849, 0.887]	—	—
accuracy	0.784 [0.757, 0.814]	—	—
specificity	0.670 [0.578, 0.761]	—	—

Confusion matrix

Table 6: S2 confusion matrix (live `c1_s2_eval`). Rows = predicted, columns = true.

Predicted	FISCAL_MEASURE	NOT_FISCAL_MEASURE
FISCAL_MEASURE	592	33
NOT_FISCAL_MEASURE	148	67

Per-act recall

Acts with passage-level recall below 1 may indicate codebook weaknesses or (more often at Tier 2) noisy labels.

Multi-measure act recall

A v0.7.0-specific secondary diagnostic: for chunks discussing multiple acts, did the model list each act?

Table 8: Per-act recall on multi-act chunks (live `c1_s2_eval$multi_measure_act_recall`).

Act	Chunks	Recalled	Recall
Changes in Depreciation Guidelines and Revenue Act of 1962	2	0	0.00
Crude Oil Windfall Profit Tax Act of 1980	2	0	0.00
Excise Tax Reduction Act of 1954	2	0	0.00
Excise Tax Reduction Act of 1965	1	0	0.00
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	4	0	0.00
Federal-Aid Highway Act of 1956	1	0	0.00
Federal-Aid Highway Act of 1959	1	0	0.00
Jobs and Growth Tax Relief Reconciliation Act of 2003	1	0	0.00
Omnibus Budget Reconciliation Act of 1987	1	0	0.00
Public Law 89-800 (Suspension of Investment Tax Credit)	2	0	0.00
Public Law 90-26 (Restoration of the Investment Tax Credit)	3	0	0.00
Revenue Act of 1948	1	0	0.00
Revenue Act of 1964	4	0	0.00
Revenue and Expenditure Control Act of 1968	2	0	0.00
Social Security Amendments of 1965	1	0	0.00
Taxpayer Relief Act of 1997 and Balanced Budget Act of 1997	1	0	0.00
Tax Reform Act of 1969	7	1	0.14
Revenue Act of 1945	5	1	0.20
Revenue Act of 1978	4	1	0.25
Tax Reduction Act of 1975	3	1	0.33

Error examples

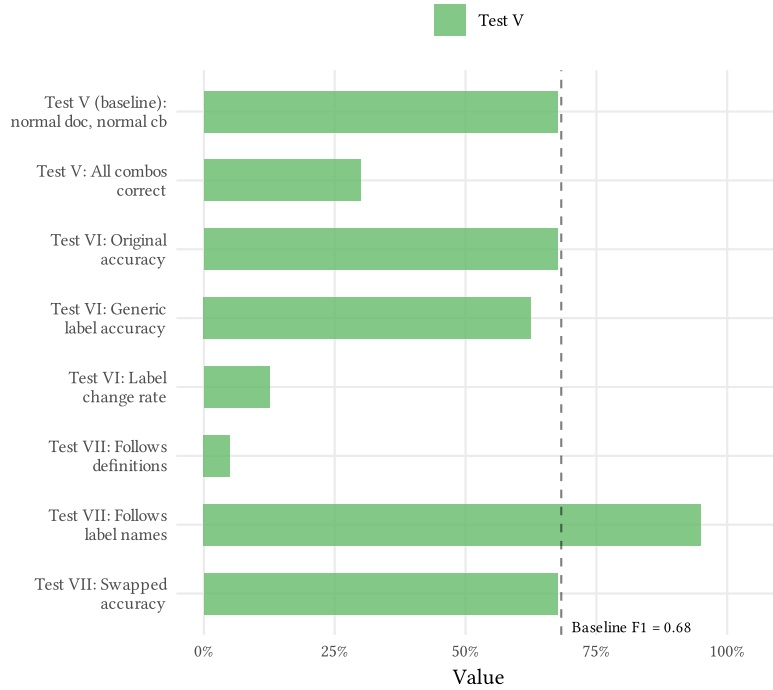
Table 9: Sample of S2 misclassifications (live `c1_s2_eval$error_analysis`).

Act	Year	Type	True	Predicted
NA	1961	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1954	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1960	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	2007	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	2002	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1980	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1973	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1967	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1992	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1978	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1974	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1974	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1986	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	2006	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE
NA	1974	negative	NOT_FISCAL_MEASURE	FISCAL_MEASURE

S3: Error Analysis (H&K Figure 4)

S3 perturbs the codebook to probe robustness: Test V (exclusion-criteria minimal pairs), Test VI (generic labels), Test VII (swapped label names).

Figure 2: S3 behavioral tests for C1 v0.7.0 (Tests V–VII), with the full-codebook baseline F1 as a reference line.



Ablation (H&K Table 4)

Components ranked by the accuracy/F1 drop when removed. Larger drops indicate more load-bearing components.

Table 10: Component ablation for the latest C1 iteration.

Condition	Accuracy	F1	Δ Accuracy	Δ F1
Full codebook	0.675	0.683	—	—
No label definitions	0.675	0.683	0.000	—
No clarifications	0.625	0.706	0.050	—
All removed	0.625	0.694	0.050	—

Multi-measure schema diagnostics

Three v0.7.0 schema checks (live [c1_s3_results](#)): are we over-listing measures, mis-tagging country of enactment, or under-listing on multi-act chunks?

Table 11: Over-listing: distribution of measures listed per FISCAL_MEASURE chunk.

Quantity	Value
Mean measures / chunk	4.50
Median (p50)	4
p90	7
Max	8
Chunks with ≥ 3 measures	6
FISCAL_MEASURE chunks	8

Table 12: Country-of-enactment tag distribution across listed measures.

Quantity	Value
Domestic (US) measures	89
OTHER (foreign) measures	0
Chunks containing any OTHER measure	0

Table 13: Under-listing: per-act recall on chunks expected to name multiple acts.

Quantity	Value
Multi-act chunks tested	3
Mean per-act recall	0.17
Chunks with full recall	0

Manual Error Analysis (H&K Table 5)

The actual S3 gate is a human read of all 40 baseline chunks against the H&K six-category framework (A–F). This is the headline result for v0.7.0.

Table 14: Manual error-analysis category distribution (latest C1 iteration).

Category	Count	Proportion
A: LLM correct	27	0.68
B: Incorrect gold standard	7	0.17
C: Document error	0	0.00
D: LLM non-compliance	0	0.00
E: Semantics/reasoning mistake	2	0.05
F: Other	4	0.10

After excluding Category B (incorrect gold standard), the label-noise-adjusted performance is:

Table 15: Bias-corrected gate metrics (Category B excluded).

Quantity	Value
Effective N	33
True positives	14
True negatives	13
False positives	4
False negatives	2
Accuracy	81.8%
Precision	77.8%
Recall	87.5%
Tier 1 recall	80.0%
Tier 2 recall	100.0%
Specificity	76.5%

The two Tier 1 false negatives in v0.7.0 (a Category E pair) are genuine semantic errors: the model recognized the relevant material but applied a stricter “substantive” bar than the codebook authorizes. The hypothesized cause is that the multi-measure schema raises the implicit threshold for what counts as a measure worth listing. The gaps were judged non-blocking and recorded for a future C1 revision; v0.7.0 was accepted for deployment.

Part 2: Development History

Everything below is sourced from `prompts/iterations/c1.yml` via the parsed `iteration_logs` target, so editing the log re-renders these figures and the narrative.

Timeline

Table 16: C1 iteration timeline (all runs, including exploratory non-Claude models).

Iter	Version	Stage	Date	Model	Pass
1	0.1.0	S1	2026-02-20	claude-haiku-4-5	✗
2	0.1.0	S1	2026-02-20	claude-haiku-4-5	✗
3	0.1.0	S1	2026-02-21	claude-haiku-4-5	✓
4	0.2.0	S1	2026-02-28	qwen/ qwen-2.5-72b-in-struct	✗
5	0.2.0	S1	2026-03-02	qwen/ qwen-2.5-72b-in-struct	✗
6	0.2.0	S2	2026-03-03	qwen/ qwen-2.5-72b-in-struct	✗
7	0.2.0	S2	2026-03-03	meta-llama/ llama-3.3-70b-in-struct	✗
8	0.2.0	S1	2026-03-03	claude-haiku-4-5	✓
9	0.2.0	S2	2026-03-03	claude-haiku-4-5	✗
10	0.3.0	S1	2026-03-03	claude-haiku-4-5	✓
11	0.3.0	S2	2026-03-03	claude-haiku-4-5	✗
12	0.4.0	S2	2026-03-04	claude-haiku-4-5	✗
13	0.4.0	S3	2026-03-09	meta-llama/ llama-3.3-70b-in-struct	—
14	0.4.0	S3	2026-03-09	meta-llama/ llama-3.3-70b-in-struct	—
15	0.4.0	S3	2026-03-09	claude-haiku-4-5	—
16	0.4.0 S3_MANUAL_ANALYSIS	S3	2026-03-10	claude-haiku-4-5	—
17	0.5.0	S1	2026-04-04	claude-haiku-4-5	✓
18	0.5.0	S2	2026-04-04	claude-haiku-4-5	✗
19	0.5.0	S3	2026-04-04	claude-haiku-4-5	—
20	0.5.0 S3_MANUAL_ANALYSIS	S3	2026-04-04	claude-haiku-4-5	—
21	0.6.0	S1	2026-04-05	claude-haiku-4-5	✓
22	0.6.0	S1	2026-04-05	google/gem- ini-2.5-flash	✗
23	0.6.0	S1	2026-04-05	google/gem- ini-2.5-flash	✗
24	0.6.0 S3_MANUAL_ANALYSIS	S3	2026-05-09	claude-haiku-4-5 ini-2.5-flash	✗

Performance trajectory

The S2 zero-shot trajectory across formal (Claude) iterations. The decisive jump came with the v0.3.0 “detection reframe” (recall 29% $\rightarrow$ 81%), and the v0.4.0 relevance-filter reframe pushed Tier 1 recall to ceiling.

Figure 3: C1 Tier 1 recall across formal iterations.

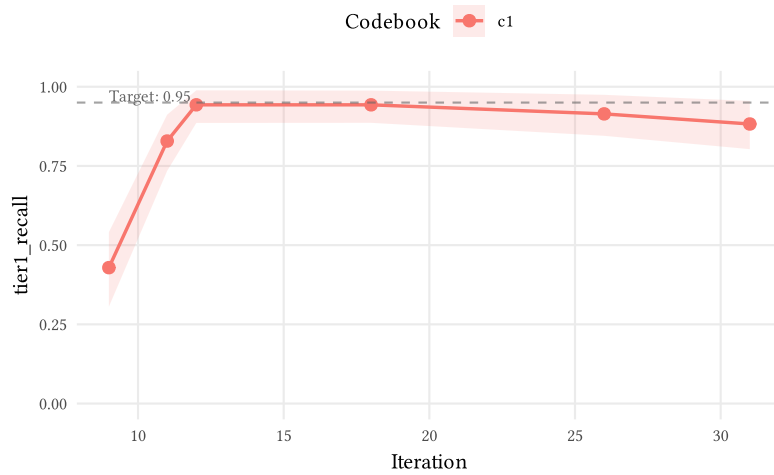
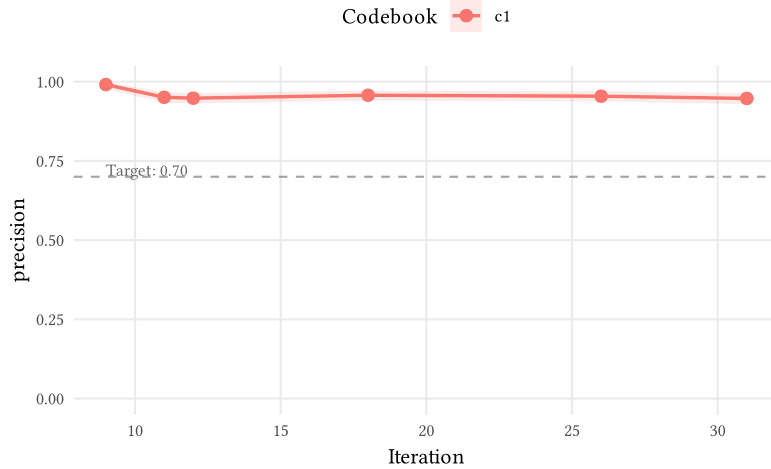


Figure 4: C1 combined recall across formal iterations.



Figure 5: C1 precision across formal iterations.



What we learned, by version

For each codebook version, the concluding iteration’s interpretation and decision (verbatim from the log).

v0.1.0 — concluded at iteration 3 (S1, 2026-02-21)

What we found. C1 passed all S1 behavioral tests. The codebook example fix was the root cause of the Test III failure — the old negative example had reasoning that contradicted its structural label, and the model followed the reasoning. Test IV also now passes (0% change rate vs 10% before), likely because the tightened NE-1 reasoning removed the hedge language that made the model order-sensitive on the Budget Enforcement Act passage.

Decision. Proceed to S2 (LOOCV evaluation).

v0.2.0 — concluded at iteration 9 (S2, 2026-03-03)

What we found. First formal S2 zero-shot evaluation with Haiku on codebook v0.2.0. Precision is near-perfect (99.1%) but recall is catastrophically low (29.1%, target $\geq 90\%$). The model applies a “dominant topic” heuristic rather than scanning for fiscal measures within chunks. 541/543 errors are false negatives with consistent reasoning patterns: “economic analysis” (27%), “general economic” (23%), “does not describe” (21%), “projections” (18%). Even Tier 1 chunks (verbatim passage matches) achieve only 42.9% recall — the model acknowledges fiscal measures exist in the text but dismisses them because the chunk’s overall character is economic analysis. 120/238 act-year rows have 0% recall (114 are Tier 2 only). The 2 false positives appear to be mislabeled negatives (Treasury Annual Reports containing genuine fiscal measures). Root cause: H&K Error Category B (prompt construction) — the codebook’s “substantive detail” requirement and negative clarifications give the model license to dismiss chunks where the fiscal measure is a minority of the content. Formal S3 skipped for this iteration because the failure mode is unambiguous and ablation would rank components of a codebook being retired.

Decision. Revise codebook to v0.3.0 with two targeted changes, then re-run S1 and S2. (1) Reframe instructions from topic classification to detection. Change the instructions to explicitly say: “Scan the entire chunk for any portion that substantively describes an enacted fiscal measure. A chunk should be classified as FISCAL_MEASURE even if the majority of the text discusses other economic topics – one substantive description anywhere in the chunk is sufficient.” (2) Soften the “substantive detail” bar but keep it meaningful. Instead of requiring “specific rate changes, affected populations, revenue estimates” (which the model interprets as a very high bar), say something like: “The passage names a specific enacted act or executive action and provides enough context to understand what fiscal liabilities it changed – this could include rate changes, affected populations, revenue estimates, effective dates, or even a general description of the measure’s purpose and scope.” This lets Tier 2 chunks (which name the act and describe it generally) pass without collapsing the bar to mere mentions. Run S3 only if v0.3.0 S2 results are in the right ballpark but need refinement.

v0.3.0 — concluded at iteration 11 (S2, 2026-03-03)

What we found. v0.3.0 detection reframe massively improved recall (+52 pp combined, +40 pp Tier 1) vs v0.2.0 (iteration 9) while keeping precision high (95.1%). The dominant-topic heuristic that caused catastrophic recall in v0.2.0 is largely resolved. Remaining 144 FN are dominated by “not enacted/proposed” reasoning (87% of FN cite this) — the model over-applies the negative clarification about unenacted proposals to chunks where enacted acts are discussed in forward-looking or conditional tense. 32 FP are largely H&K Error Category F (ambiguous ground truth): genuine fiscal measures not in the 44-act label set, meaning reported precision is a conservative lower bound. 19 zero-recall act-year rows are all Tier 2-only single-chunk distant references. Tier 1 and Tier 2 miss rates are similar (~17-19%), suggesting the problem is the model’s enacted-status threshold, not reference vagueness. 0 parse failures (100% valid JSON from Haiku).

Decision. Revise codebook to v0.4.0 with a strategic reframe based on discussion of results. Three changes: (1) Remove the enacted/proposed filter from C1 entirely — C1 becomes a recall-optimized relevance filter for fiscal policy discussion; enacted status determination moves downstream to a future filter or C2. If we ex post find more acts than R&R, we add an enacted filter downstream. (2) Further soften the substantive detail bar to push Tier 1 recall toward 95%+; Tier 1 recall is the top priority because these chunks carry the richest information for C2-C4 classification. (3) Accept that reported precision is conservative due to Category F FPs in the negative test set. Precision cost is bearable because downstream codebooks (C2-C4) will filter irrelevant chunks. Additionally, improve the act identification pipeline to search for acronym mentions of known acts (e.g., ERTA, OBRA, TRA), improving label coverage for evaluation.

v0.4.0 — concluded at iteration 16 (S3_MANUAL_ANALYSIS, 2026-03-10)

What we found. There may be room for codebook improvement. The 3 Category E errors share a common pattern: the model conflates government

financial actions (intergovernmental transfers, education appropriations, debt management) with taxpayer-liability-changing fiscal measures. The Category F case (EIA) reveals R&R boundary ambiguity on organizational acts with fiscal implications. The 5 Category B errors confirm that all false negatives are gold-standard noise from IKA Tier 2 matching, not codebook or model failures. Bias-corrected recall is 100%.

Decision. Consider improving the codebook with a clarification distinguishing changes to taxpayers' liabilities from other government financial actions (intergovernmental transfers, debt management, organizational acts). This is the single highest-leverage codebook improvement identified by manual analysis.

v0.5.0 — concluded at iteration 20 (S3_MANUAL_ANALYSIS, 2026-04-04)

What we found. C1 v0.5.0 achieves zero semantic/reasoning errors (Category E = 0), up from 3 in v0.4.0. The taxpayer-liability clarification added in v0.5.0 fixed text 35 (TIPS). Bias-corrected recall is 100% across both tiers. All residual errors are Category F (codebook scope ambiguity): 3 false positives where the model correctly identified real fiscal content that falls outside R&R's tax/revenue scope. These share a common root cause — C1's definition of 'fiscal measure' does not encode R&R's implicit boundary between tax legislation and spending-side policy. A deeper insight: some of these measures live in a gray zone where spending policy is implemented through tax-adjacent mechanisms (credits, rebates), and their exclusion from R&R may be driven by motivation-classification needs in C2 rather than C1 scope alone. This is a potential source of C1 improvement, either through codebook clarification or pipeline design (downstream C2 filter). With n=3, this is a hypothesis to monitor during C2 development, not a confirmed systematic pattern.

Decision. C1 S3 gate passed. Proceed to C2 (Motivation) development. During C2 development, monitor the tax-vs-spending gray zone: observe how measures with behavioral/political motivation (energy policy, welfare reform) interact with R&R's 4-bucket classification. This will inform whether the F-error pattern requires a C1 codebook fix or is better addressed as a C2-level filter.

v0.6.0 — concluded at iteration 28 (S3_MANUAL_ANALYSIS, 2026-04-07)

What we found. C1 v0.6.0 S3 regression check complete. Zero semantic errors (E=0), unchanged from v0.5.0. The single new prediction (text_id 24) is a positive finding: the extra_output_fields prompted the model to read footnotes more carefully, correctly identifying two R&R-relevant fiscal measures (SS Amendments 1950, Revenue Act 1951) that the v0.5.0 model and iteration 20's human reviewer both overlooked. Bias-corrected recall is 100% across both tiers. The 3 residual F errors (texts 23, 32, 33) remain the same spending-side scope ambiguity from v0.5.0 — these are deferred to C2 development as planned in iteration 20.

Decision. C1 S3 gate passed for v0.6.0. Proceed to C2 (Motivation) codebook development. The extra_output_fields are a net positive — they improve model thoroughness without introducing new errors. During C2 development, monitor the 3 F-error pattern (spending-side fiscal policy) as planned.

v0.7.0 — concluded at iteration 33 (S3_MANUAL_ANALYSIS, 2026-05-20)

What we found. New schema either changes the model’s classification framework or distracts it from reasoning the main codebook task (fiscal shock identification). The 2 new Tier 1 E errors (text_ids 3, 7) are both cases where the model recognized the relevant content and then applied an overly restrictive bar — consistent with the hypothesis that asking for per-measure structure (name + country + 3 relevance flags) raises the model’s threshold for what counts as a measure worth listing. Tier 2 recall remains at ceiling (100%) after excluding IKA noise; specificity drops modestly because the codebook-gap F-category continues to flag substantive foreign-credit and spending-side content as FISCAL_MEASURE. Net: v0.7.0 is fit-for-purpose for Phase 2 deployment despite the Tier 1 regression — the multi-measure schema unlocks the per-measure structure required for C2a (Phase 5) and Malaysia (Phase 2) downstream consumption.

Decision. Defer identified gap fixes to a future C1 version. Accept v0.7.0 as-is for deployment. Three gaps recorded for future revision (foreign-credit/capital-controls, spending-side authorizations, overly-restrictive substantive threshold) but not blocking. C1 S3 manual audit closes Step 4 of the 2026-05-18 sequencing plan in docs/deltas.md. Next: Step 5 (C2a v0.6.0).